*****To Setup GitHub Actions CI & CD with Docker Server on Amazon Linux 2023*****

Step 1:- Launch 2 EC2 instance in mumbai region (ap-south-1}
Instance Name:- 1) Git_Server
                2) Docker_Server

Instance Type:- t3.micro
Key:- github_actions_key
create .pem
VPC:- Default
SG:- Open-ssh-http-icmp-cutom-tcp
range:- 8080-9090
Storage:- 20 GB
Launch Instance

Step 2:- Login to git server
and run below command

```
    #sudo su -
    #hostnamectl set-hostname gitsrv.example.com
    #hostname
    #yum install git -y
#git --version     {to check git version}
```

Step 3:- Create Project

```
#mkdir /root/docker-ec2-deploy
    #cd docker-ec2-deploy
    #mkdir src
    #mkdir -p .github/workflows
```

Step 4:- Application Creation

```
#vim package.json
{
  "name": "mywebapp",
  "version": "1.0.0",
  "main": "src/index.js",
  "scripts": {
    "start": "node src/index.js"
  },
  "dependencies": {
    "express": "^4.18.2"
  }
}

:wq!   ---------->{forcefully save and quite}

       #vim src/index.js
const express = require("express");

const app = express();

const PORT = 80;

app.get("/", (req,res)=>{
 res.send("GitHub Actions Docker Deployment Successful");
});


app.listen(PORT,()=>{
 console.log("Server running on port 80");
});
```

Step 5:- Create Dockerfile

```
#vim Dockerfile
```

```
FROM node:18-alpine

WORKDIR /usr/src/app

COPY package*.json ./

RUN npm install

COPY . .

EXPOSE 80

CMD ["npm","start"]
```

Step 6:- GitHub Actions Workflow
#vim .github/workflows/deploy.yml

```
name: Deploy Docker WebApp to EC2

on:
  push:
    branches:
      - main


jobs:

 deploy:

  runs-on: ubuntu-latest


  steps:

  - name: Checkout Code
    uses: actions/checkout@v4


  - name: Build Docker Image
    run: |
      docker build -t mywebapp:latest .


  - name: Save Docker Image
    run: |
      docker save mywebapp:latest -o mywebapp.tar
      chmod 644 mywebapp.tar


  - name: Copy Image to Docker Server
    uses: appleboy/scp-action@v0.1.7

    with:

      host: ${{ secrets.EC2_HOST }}

      username: ${{ secrets.EC2_USER }}

      key: ${{ secrets.EC2_SSH_KEY }}

      source: mywebapp.tar

      target: /home/ec2-user/
```

```
  - name: Deploy Container

    uses: appleboy/ssh-action@v1.0.3

    with:

      host: ${{ secrets.EC2_HOST }}

      username: ${{ secrets.EC2_USER }}

      key: ${{ secrets.EC2_SSH_KEY }}


      script: |

        sudo docker load -i /home/ec2-user/mywebapp.tar

        sudo docker stop mywebapp || true

        sudo docker rm mywebapp || true

        sudo docker run -d \
        --name mywebapp \
        -p 8086:80 \
        mywebapp:latest


:wq!   ---------->{forcefully save and quite}


Step 7:- Docker Server Setup
Login to docker server
once docker server login do not swith to root user
we all steps perform on ec2-user
        $sudo hostnamectl set-hostname dockersrv.example.com
            $hostname
        $sudo dnf install docker -y
            $sudo systemctl enable --now docker
            $docker --version


Step 8:- SSH Key Setup on docker server
        $ssh-keygen -t rsa
            Public key:
            $cat ~/.ssh/id_rsa.pub
            $cd .ssh
Add:    $vi ~/.ssh/authorized_keys
Permission:   $chmod 700 ~/.ssh
              $chmod 600 ~/.ssh/authorized_keys
Private key:
        $cat ~/.ssh/id_rsa
Copy to GitHub Secret.


Step 9:- Now go to github account
Create new repository
name:- docker-ec2-deploy
Create repository
and copy 3 newly command


Step 10:- Create GitHub Secrets
select docker-ec2-deploy repository
go to Settings->Secrets and variables->Actions
```

Create:

Name	Value
EC2_HOST	Docker Server Public IP
EC2_USER	ec2-user
EC2_SSH_KEY	id_rsa private key

Now go to Git server
Step 11:- Create Git initialize
```
#cd docker-ec2-deploy
#git init
```

Step 12:- Git user configure
```
#git config --global user.name "Sai"
#git config --global user.email "sai3cs@gmail.com"
```

Step 13:- add file and commit
```
#git add .
#git commit -m "Initial Docker Project"
```

Step 14:- To conect GitHub repository
```
#git remote add origin https://github.com/username/docker-ec2-deploy.git
```

Step 15:- to Push git code
```
#git branch -M main
#git push -u origin main
```
Username:-
Password

Step 16:- Go to github dashboard
Click on docker-ec2-deploy-->.github/workflows--->Actions--->Select Running Job
once job is successfully completed

Step 17:- Go to Docker Server
and run below command via ec2-user
```
$sudo docker image
```

Expected:
```
REPOSITORY     TAG
mywebapp       latest
```

Check running container:
```
$sudo docker ps
```

Expected:
```
CONTAINER ID   IMAGE          PORT
xxxxx          mywebapp       0.0.0.0:8086->80/tcp
```

Step 18:- Test Application from Browser
Open any Browser
http://Docker_Server_Public_IP:8086

Output:

GitHub Actions Docker Deployment Successful

***********************Thank You!********************************